TUGAS PBO (PEMROGRAMAN BERORIENTASI OBJEK)

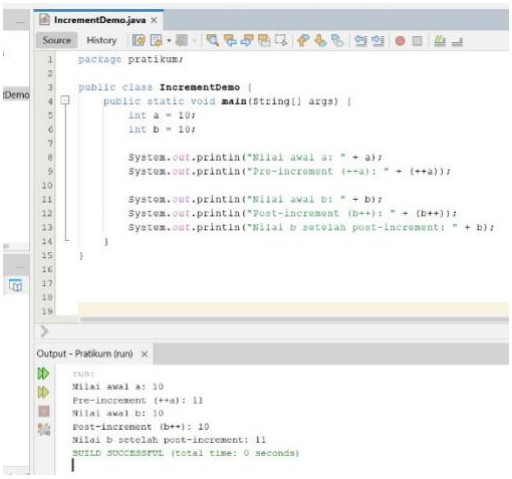
Nama :

Kelas :

NPM :

SHIFT :

1. **Program Ilustrasi Operator Increment**



```
package pratikum;

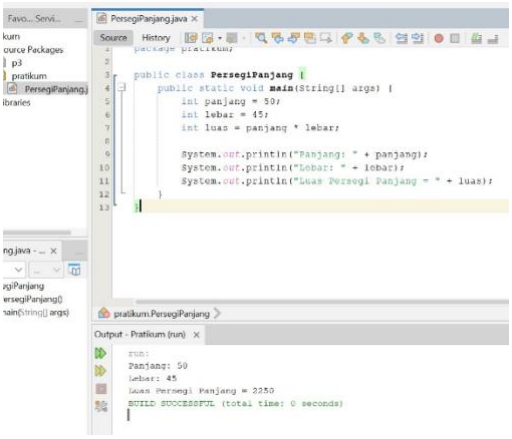
public class IncrementDemo {
    public static void main(String[] args) {
        int a = 10;
        int b = 10;

        System.out.println("Nilai awal a: " + a);
        System.out.println("Pre-increment (++a): " + (++a));

        System.out.println("Nilai awal b: " + b);
        System.out.println("Post-increment (b++): " + (b++));
        System.out.println("Nilai b setelah post-increment: " + b);
    }
}
```

```
run:
Nilai awal a: 10
Pre-increment (++a): 11
Nilai awal b: 10
Post-increment (b++): 10
Nilai b setelah post-increment: 11
BUILD SUCCESSFUL (total time: 0 seconds)
```

2. **Program Luas Persegi Panjang**

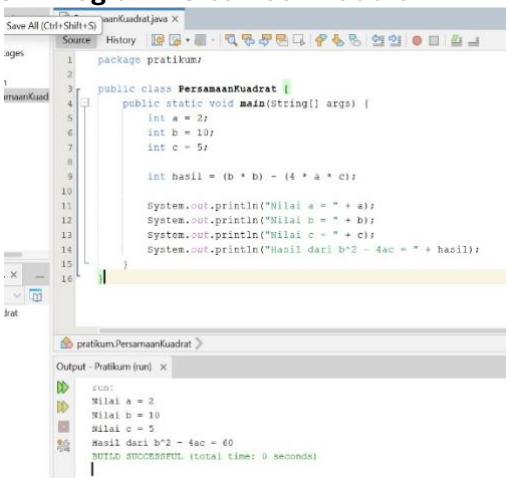


```
public class PersegiPanjang {
    public static void main(String[] args) {
        int panjang = 50;
        int lebar = 45;
        int luas = panjang * lebar;

        System.out.println("Panjang: " + panjang);
        System.out.println("Lebar: " + lebar);
        System.out.println("Luas Persegi Panjang = " + luas);
    }
}
```

```
run:
Panjang: 50
Lebar: 45
Luas Persegi Panjang = 2250
BUILD SUCCESSFUL (total time: 0 seconds)
```

3. **Program Persamaan Kuadrat**



```
package pratikum;

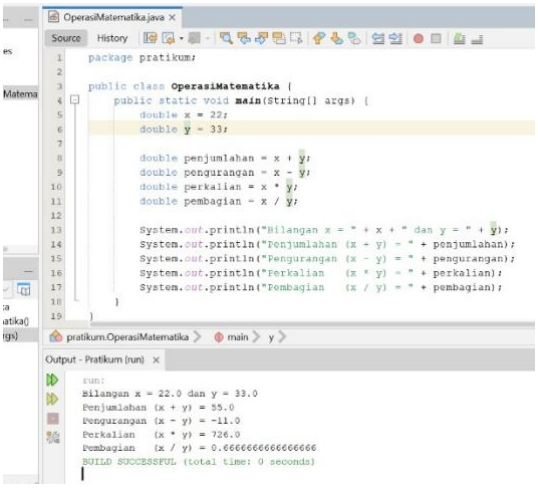
public class PersamaanKuadrat {
    public static void main(String[] args) {
        int a = 2;
        int b = 10;
        int c = 5;

        int hasil = (b * b) - (4 * a * c);

        System.out.println("Nilai a = " + a);
        System.out.println("Nilai b = " + b);
        System.out.println("Nilai c = " + c);
        System.out.println("Hasil dari b^2 - 4ac = " + hasil);
    }
}
```

```
run:
Nilai a = 2
Nilai b = 10
Nilai c = 5
Hasil dari b^2 - 4ac = 60
BUILD SUCCESSFUL (total time: 0 seconds)
```

4. Program Operasi Matematika



```
package pratikum;

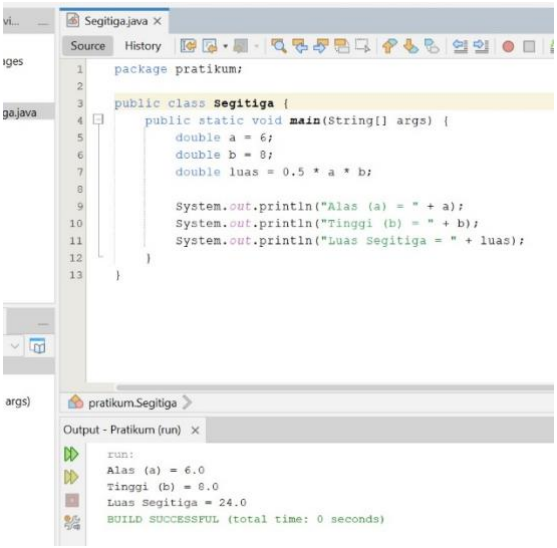
public class OperasiMatematika {
    public static void main(String[] args) {
        double x = 22;
        double y = 33;

        double penjumlahan = x + y;
        double pengurangan = x - y;
        double perkalian = x * y;
        double pembagian = x / y;

        System.out.println("Bilangan x = " + x + " dan y = " + y);
        System.out.println("Penjumlahan (x + y) = " + penjumlahan);
        System.out.println("Pengurangan (x - y) = " + pengurangan);
        System.out.println("Perkalian   (x * y) = " + perkalian);
        System.out.println("Pembagian   (x / y) = " + pembagian);
    }
}
```

```
run:
Bilangan x = 22.0 dan y = 33.0
Penjumlahan (x + y) = 55.0
Pengurangan (x - y) = -11.0
Perkalian   (x * y) = 726.0
Pembagian   (x / y) = 0.6666666666666666
BUILD SUCCESSFUL (total time: 0 seconds)
```

5. Program Luas Segitiga



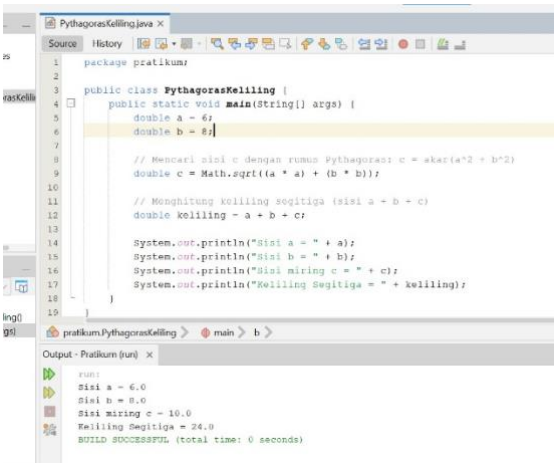
```
package pratikum;

public class Segitiga {
    public static void main(String[] args) {
        double a = 6;
        double b = 8;
        double luas = 0.5 * a * b;

        System.out.println("Alas (a) = " + a);
        System.out.println("Tinggi (b) = " + b);
        System.out.println("Luas Segitiga = " + luas);
    }
}
```

```
run:
Alas (a) = 6.0
Tinggi (b) = 8.0
Luas Segitiga = 24.0
BUILD SUCCESSFUL (total time: 0 seconds)
```

6. Program pythagoras Dan Keliling Segitiga



```
package pratikum;

public class PythagorasKeliling {
    public static void main(String[] args) {
        double a = 6;
        double b = 8;

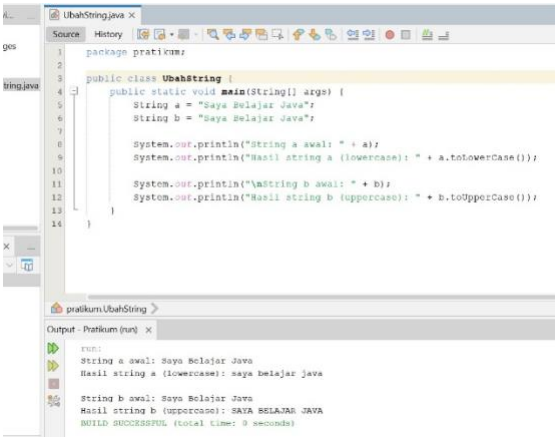
        // Mencari sisi c dengan rumus Pythagoras: c = akar(a^2 + b^2)
        double c = Math.sqrt((a * a) + (b * b));

        // Menghitung keliling segitiga (sisi a + b + c)
        double keliling = a + b + c;

        System.out.println("Sisi a = " + a);
        System.out.println("Sisi b = " + b);
        System.out.println("Sisi miring c = " + c);
        System.out.println("Keliling Segitiga = " + keliling);
    }
}
```

```
run:
Sisi a = 6.0
Sisi b = 8.0
Sisi miring c = 10.0
Keliling Segitiga = 24.0
BUILD SUCCESSFUL (total time: 0 seconds)
```

7. Program Pengubah String



```
package pratikum;

public class UbahString {
    public static void main(String[] args) {
        String a = "Saya Belajar Java";
        String b = "Saya Belajar Java";

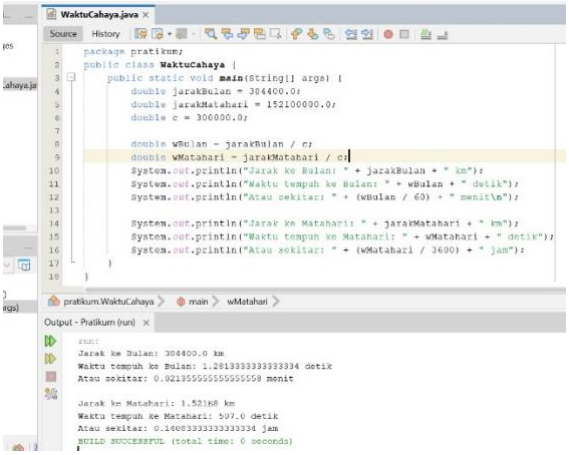
        System.out.println("String a awal: " + a);
        System.out.println("Hasil string a (lowercase): " + a.toLowerCase());

        System.out.println("\nString b awal: " + b);
        System.out.println("Hasil string b (uppercase): " + b.toUpperCase());
    }
}
```

```
run:
String a awal: Saya Belajar Java
Hasil string a (lowercase): saya belajar java

String b awal: Saya Belajar Java
Hasil string b (uppercase): SAYA BELAJAR JAVA
BUILD SUCCESSFUL (total time: 0 seconds)
```

8. Program Waktu Tempuh Cahaya



```
package pratikum;
public class WaktuCahaya {
    public static void main(String[] args) {
        double jarakBulan = 384400.0;
        double jarakMatahari = 152100000.0;
        double c = 300000.0;

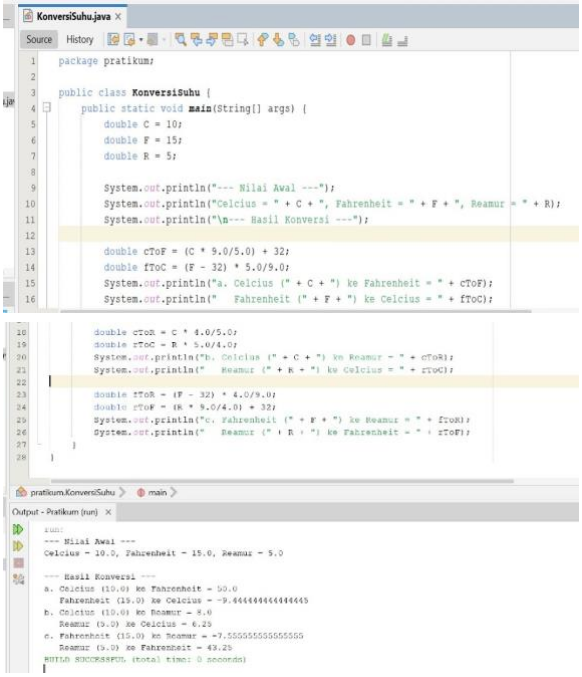
        double wBulan = jarakBulan / c;
        double wMatahari = jarakMatahari / c;
        System.out.println("Jarak ke Bulan: " + jarakBulan + " km");
        System.out.println("Waktu tempuh ke Bulan: " + wBulan + " detik");
        System.out.println("Atau sekitar: " + (wBulan / 60) + " menit\n");

        System.out.println("Jarak ke Matahari: " + jarakMatahari + " km");
        System.out.println("Waktu tempuh ke Matahari: " + wMatahari + " detik");
        System.out.println("Atau sekitar: " + (wMatahari / 3600) + " jam");
    }
}
```

```
run:
Jarak ke Bulan: 384400.0 km
Waktu tempuh ke Bulan: 1.2813333333333334 detik
Atau sekitar: 0.021355555555555558 menit

Jarak ke Matahari: 1.52168 km
Waktu tempuh ke Matahari: 507.0 detik
Atau sekitar: 0.14083333333333334 jam
BUILD SUCCESSFUL (total time: 0 seconds)
```

9. Program Konvesi Suhu



```
package pratikum;

public class KonversiSuhu {
    public static void main(String[] args) {
        double C = 10;
        double F = 15;
        double R = 5;

        System.out.println("--- Nilai Awal ---");
        System.out.println("Celcius = " + C + ", Fahrenheit = " + F + ", Reamur = " + R);
        System.out.println("\n--- Hasil Konversi ---");

        double cToF = (C * 9.0/5.0) + 32;
        double fToC = (F - 32) * 5.0/9.0;
        System.out.println("a. Celcius (" + C + ") ke Fahrenheit = " + cToF);
        System.out.println("   Fahrenheit (" + F + ") ke Celcius = " + fToC);

        double cToR = C * 4.0/5.0;
        double rToC = R * 5.0/4.0;
        System.out.println("b. Celcius (" + C + ") ke Reamur = " + cToR);
        System.out.println("   Reamur (" + R + ") ke Celcius = " + rToC);

        double fToR = (F - 32) * 4.0/9.0;
        double rToF = (R * 9.0/4.0) + 32;
        System.out.println("c. Fahrenheit (" + F + ") ke Reamur = " + fToR);
        System.out.println("   Reamur (" + R + ") ke Fahrenheit = " + rToF);
    }
}
```

```
run:
--- Nilai Awal ---
Celcius = 10.0, Fahrenheit = 15.0, Reamur = 5.0

--- Hasil Konversi ---
a. Celcius (10.0) ke Fahrenheit = 50.0
   Fahrenheit (15.0) ke Celcius = -9.444444444444445
b. Celcius (10.0) ke Reamur = 8.0
   Reamur (5.0) ke Celcius = 6.25
c. Fahrenheit (15.0) ke Reamur = -7.555555555555555
   Reamur (5.0) ke Fahrenheit = 43.25
BUILD SUCCESSFUL (total time: 0 seconds)
```